

Hydralazine: Drug information - UpToDate

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For additional information see "Hydralazine: Patient drug information" and "Hydralazine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: Canada

- APO-HydRALAZINE;
- Apresoline;
- JAMP-HydRALAZINE;
- MINT-HydRALAZINE

Pharmacologic Category

- Antihypertensive;
- Vasodilator

Dosing: Adult

Heart failure with reduced ejection fraction

Heart failure with reduced ejection fraction (off-label use):

Note: Alternative therapy for patients with persistent NYHA class III or IV heart failure with reduced ejection fraction (HFrEF) who cannot tolerate an angiotensin II receptor-neprilysin inhibitor, angiotensin-converting enzyme inhibitor, or angiotensin II receptor blocker or additional therapy in patients who have residual symptoms despite an optimal medical regimen for HFrEF (Ref).

Oral: Initial: 25 mg 3 times daily in combination with isosorbide dinitrate; titrate dose as tolerated every ≥ 1 to 2 weeks; target dose: 75 or 100 mg 3 times daily (Ref).

Hypertension, chronic

Hypertension, chronic (alternative agent):

Note: Not recommended for initial management but may be considered as additional therapy for resistant hypertension in patients who do not respond adequately to combination therapy with preferred agents. Consider combining with a beta-blocker and/or diuretic because hydralazine is associated with reflex tachycardia and fluid retention (Ref). For severe asymptomatic hypertension, may consider short-term use for BP lowering (eg, over hours) if there is concern that severe BP elevation will precipitate an acute cardiovascular event, such as in patients with known aortic or intracranial aneurysms. Some experts recommend captopril or clonidine, and hydralazine is considered an alternative agent in this situation (Ref).

Oral: Initial: 10 mg 4 times daily for 2 to 4 days, then 25 mg 4 times daily for the remainder of the week followed by titration based on response to 50 mg 4 times daily (Ref); usual dosage range: 100 to 200 mg/day in divided doses (Ref). Maximum dose: 300 mg/day in divided doses (Ref); however, doses >200 mg/day are generally avoided due to increased risk of lupus-like reaction (Ref). Patients with severe asymptomatic hypertension and no signs of acute end organ damage should be evaluated for medication adjustment within 1 week (Ref).

Hypertensive emergency

Hypertensive emergency (alternative agent):

Note: For severe elevation in blood pressure associated with new or worsening target-organ damage. Hyalazine is generally not recommended due to unpredictable and prolonged antihypertensive effects (Ref):

IM, IV: 10 to 20 mg every 4 to 6 hours as needed; may increase dose to a maximum of 40 mg/**dose** if necessary; some experts recommend a maximum of 20 mg/**dose**; maximum cumulative dose: 200 mg per 24 hours. Consider combining with a beta-blocker since hyalazine is associated with reflex tachycardia (Ref).

Hypertensive emergency in pregnancy or postpartum

Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia) (off-label use):

Note: For acute-onset, severe, persistent hypertension (eg, systolic BP ≥ 160 mm Hg or diastolic BP ≥ 110 mm Hg) (Ref).

IV: Initial: 5 or 10 mg; repeat with 5 to 10 mg doses every 20 to 40 minutes if BP continues to exceed thresholds (Ref). If systolic BP or diastolic BP remains above threshold after a total cumulative dose of 20 to 30 mg or if heart rate exceeds 100 beats per minute, another agent should be used (Ref).

Perioperative hypertension

Perioperative hypertension (alternative agent):

Note: For patients with chronic hypertension prior to surgery, restart oral therapies as soon as appropriate once hemodynamically stable. The antihypertensive effects of hyalazine are less predictable than other parenteral antihypertensives and can cause reflex tachycardia. Some experts suggest doses in the low end of the dosing range (Ref).

IV: 5 to 20 mg every 4 to 6 hours as needed (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: There are no dosage adjustments provided in the manufacturer's labeling. Limited data suggest concentrations of hydralazine and/or its metabolites are increased in patients with CrCl <50 mL/minute; however, results vary and there is no evidence of increased adverse effects (Ref).

Altered kidney function (Aronoff 2007; expert opinion derived from Levin 2010):

GFR ≥ 10 mL/minute: **IM, IV, Oral:** No dosage adjustment necessary.

GFR <10 mL/minute:

IM, IV: No dosage adjustment necessary when used short term; use with caution (Ref).

Oral: Administer usual dose every 8 to 12 hours (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzed (Ref).

IM, IV: No dosage adjustment necessary when used short term; use with caution (Ref).

Oral: Administer usual dose every 8 to 12 hours (Ref). Avoid administering predialysis in patients who are prone to intra-dialytic hypotension (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzed (Ref).

IM, IV: No dosage adjustment necessary when used short term; use with caution (Ref).

Oral: Administer usual dose every 8 to 12 hours (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; however, hyalazine undergoes extensive hepatic metabolism.

Dosing: Older Adult

Refer to adult dosing. Monitor and titrate carefully to reduce the risk for adverse events in older adults (Ref).

Dosing: Pediatric

(For additional information see "Hyalazine: Pediatric drug information")

Dosage guidance:

Dosing: Individualize dose; titrate to patient response.

Heart failure, afterload reduction

Heart failure, afterload reduction: Limited data available (Ref): **Note:** May cause reflex tachycardia; use in combination with beta-blocker therapy has been suggested (Ref).

IV:

Infants: 0.1 to 0.5 mg/kg/dose every 6 to 8 hours; maximum dose: 2 mg/kg/dose.

Children and Adolescents: 0.15 to 0.2 mg/kg/dose every 4 to 6 hours; maximum dose: 20 mg/dose.

Oral: Infants, Children, and Adolescents: 0.75 to 3 mg/kg/**day** divided every 6 to 12 hours; maximum daily dose: 7 mg/kg/**day** or 200 mg/**day**, whichever is less.

Hypertension, chronic

Hypertension, chronic: Children and Adolescents: Oral: Initial: 0.75 mg/kg/**day** in 2 to 4 divided doses; maximum initial adult dose: 10 mg/dose; may increase gradually over 3 to 4 weeks; maximum daily dose: 7.5 mg/kg/**day** not to exceed 200 mg/**day** (Ref).

Hypertension, acute severe

Hypertension, acute severe : Note: Oral therapy should be used in patients with less severe symptoms; IV and IM therapy should be reserved for use in patients with life-threatening symptoms (Ref): Infants, Children, and Adolescents:

Oral: 0.25 mg/kg/dose every 6 to 8 hours as needed; maximum dose: 25 mg/dose (Ref).

IM, IV: Initial: 0.1 to 0.2 mg/kg/dose every 4 to 6 hours, titrate as needed; usual dosage range: 0.2 to 0.6 mg/kg/dose every 4 to 6 hours; maximum dose: 20 mg/dose (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; however, the following adjustments have been recommended (Ref). **Note:** Renally adjusted dose recommendations are based on doses: Oral: 0.75 to 1 mg/kg/**day** divided every 6 to 12 hours; maximum daily dose: 200 mg/**day**; IV: 0.1 to 0.2 mg/kg/dose every 6 hours; maximum dose: 20 mg/dose.

GFR >50 mL/minute/1.73 m²: No adjustment necessary.

GFR 10 to 50 mL/minute/1.73 m²: Administer every 8 hours.

GFR <10 mL/minute/1.73 m²: Administer every 12 to 24 hours.

Intermittent hemodialysis: Administer every 12 to 24 hours.

Peritoneal dialysis (PD): Administer every 12 to 24 hours.

Continuous renal replacement therapy (CRRT): Administer every 8 hours.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; however, hyalalazine undergoes extensive hepatic metabolism.

Adverse Reactions (Significant): Considerations

Lupus-like syndrome

Hyalalazine-induced **lupus-like syndrome** (HILS) has occurred in up to 10% of patients on doses ≥ 200 mg for ≥ 3 months (Ref) and is reversible upon discontinuation. HILS commonly results in nonadherence or discontinuation and may lead to multi-organ involvement (pericarditis, hepatosplenomegaly); severe malaise/myalgias; acute joint pain and swelling; blood dyscrasias (leukopenia, thrombocytopenia); and positive serologies (antinuclear antibodies, antihistone antibodies, elevated erythrocyte sedimentation rate) (Ref).

Mechanism: Exact mechanism is unknown; may occur via IL-10 suppression at higher doses of hyalalazine (Ref).

Onset: Delayed; onset reported after at least 3 months of therapy. However, this may vary based on dosing regimen (Ref). Duration of therapy in case reports varies from 1 to 24 months (Ref).

Risk factors:

- Females (Ref)
- White race (Ref)
- Slow hepatic acetylation (Ref)
- Doses ≥ 200 mg per day (Ref)
- Cumulative dose >100 g (Ref)
- Treatment for at least 3 months (Ref)
- Human leukocyte antigen DR4 genotype (Ref)
- Family history of autoimmune disease (Ref)
- History of thyroid disease (Ref)
- Null gene for C4 (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Acute myocardial infarction, angina pectoris, edema (Ref), flushing, hypotension, myocardial stimulation, palpitations, paradoxical response to antihypertensive, tachycardia (Ref)

Dermatologic: Pruritus, skin rash (including eczema) (Ref), urticaria (Ref)

Gastrointestinal: Anorexia (Ref), constipation, diarrhea (Ref), nausea (Ref), paralytic ileus, vomiting (Ref)

Genitourinary: Difficulty in micturition

Hematologic & oncologic: Agranulocytosis (Ref), eosinophilia, leukopenia, lymphadenopathy, purpuric disease, splenomegaly

Hepatic: Hepatitis (Ref)

Nervous system: Chills, dizziness, headache (Ref), peripheral neuropathy (Ref), psychotic reaction (including anxiety, depression, disorientation, euphoria, hypomania, nervousness) (Ref)

Neuromuscular & skeletal: Arthralgia, muscle cramps, tremor

Ophthalmic: Conjunctivitis, lacrimation

Respiratory: Dyspnea, nasal congestion

Miscellaneous: Fever (Ref)

Postmarketing:

Cardiovascular: Vasculitis (Ref)

Dermatologic: Toxic epidermal necrolysis (Ref)

Hematologic & oncologic: Hemolytic anemia (Ref)

Hepatic: Hepatotoxicity (Ref)

Nervous system: Fatigue (Ref), malaise (Ref)

Neuromuscular & skeletal: Lupus-like syndrome (common: $\geq 10\%$ in doses ≥ 200 mg for ≥ 3 months (Ref); multi-organ involvement [eg, pericarditis, hepatosplenomegaly], severe malaise/myalgias, acute joint pain and swelling, blood dyscrasias [eg, leukopenia, thrombocytopenia], positive serologies [eg, antinuclear antibodies, antihistone antibodies, elevated erythrocyte sedimentation rate] (Ref), and lupus erythematosus-like rash (Ref))

Renal: Glomerulonephritis (Ref)

Contraindications

Hypersensitivity to hydralazine or any component of the formulation; coronary artery disease; mitral valve rheumatic heart disease

Canadian labeling: Additional contraindications (not in US labeling): Idiopathic systemic lupus erythematosus and related diseases; severe tachycardia and heart failure (HF) with high cardiac output (eg, in thyrotoxicosis); myocardial insufficiency due to mechanical obstruction (eg, aortic or mitral stenosis, constrictive pericarditis); isolated right ventricular HF due to pulmonary hypertension; acute dissecting aneurysm of the aorta; porphyria (injection only)

Warnings/Precautions

Disease-related concerns:

- Cardiovascular disease: Use is contraindicated in patients with coronary artery disease (CAD). Use with caution in patients with cerebral vascular accidents and suspected CAD; myocardial stimulation produced by hydralazine can cause anginal attacks and electrocardiogram (ECG) changes of myocardial ischemia; has been implicated in the production of myocardial infarction. According to the American Heart Association/American College of Cardiology/American Society of Hypertension 2015 scientific statement for the treatment of hypertension in patients with CAD, hydralazine (without a concomitant nitrate [eg, isosorbide dinitrate]) should be avoided for the treatment of hypertension in patients with heart failure (with reduced ejection fraction) of ischemic origin (AHA/ACC/ASH [Rosendorff 2015]).

- Kidney impairment: Use with caution in patients with advanced kidney impairment; dosage adjustment recommended.

- Mitral valvular disease: Use with caution in patients with mitral valvular disease; may increase pulmonary artery pressure in these patients. Use is contraindicated in patients with mitral valve rheumatic heart disease.

Dosage form specific issues:

- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).
- Tartrazine sensitivity: May contain tartrazine, which may cause allergic-type reactions (including bronchial asthma) in certain susceptible individuals. Although the overall incidence of tartrazine sensitivity in the general population is low, it is frequently seen in patients who are also hypersensitive to aspirin.

Warnings: Additional Pediatric Considerations

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Generic: 20 mg/mL (1 mL)

Solution, Injection, as hydrochloride [preservative free]:

Generic: 20 mg/mL (1 mL)

Tablet, Oral, as hydrochloride:

Generic: 10 mg, 25 mg, 50 mg, 100 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (hyarALAZINE HCl Injection)

20 mg/mL (per mL): \$4.20 - \$20.45

Tablets (hyarALAZINE HCl Oral)

10 mg (per each): \$0.26 - \$0.41

25 mg (per each): \$0.07 - \$0.51

50 mg (per each): \$0.08 - \$0.56

100 mg (per each): \$0.09 - \$1.01

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect,

incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Apresoline: 20 mg/mL (1 mL)

Generic: 20 mg/mL (1 mL)

Tablet, Oral, as hydrochloride:

Generic: 10 mg, 25 mg, 50 mg

Extemporaneous Preparations

4 mg/mL Oral Suspension (ASHP standard concentration) (Ref)

A 4 mg/mL oral suspension may be made with tablets and a 1:1 mixture of Ora-Sweet SF and Ora-Plus. Crush four 100 mg tablets in a mortar and reduce to a fine powder. Add 15 mL of the vehicle and mix to a uniform paste; mix while adding the vehicle in incremental proportions to **almost** 100 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of vehicle sufficient to make 100 mL. Label "shake well", "protect from light", "store in a refrigerator." Stable for 2 days when stored in amber plastic prescription bottles in the dark and refrigerated (Ref).

Note: Stability reduced to 24 hours if Ora-Sweet is substituted for Ora-Sweet SF.

Administration: Adult

Oral: Administer without regard to meals. However, food enhances bioavailability; administer consistently with regard to meals.

Injection: Response may be delayed and unpredictable in some patients; titrate cautiously to response.

IM: Administer undiluted as IM injection.

IV: Administer undiluted as slow IV push.

Preparation for Administration: Adult

Parenteral: IV: Manufacturer does not recommend dilution. If further dilution is desired hydralazine should be diluted in NS for IVPB administration due to decreased stability in D5W; hydralazine diluted in NS to a concentration of 0.2 mg/mL is reported as stable for at least 1 day at room temperature (Ref).

Administration: Pediatric

Oral: May administer without regard to meals. However, food enhances bioavailability; administer consistently with regard to meals.

Parenteral:

IM: Administer undiluted as IM injection.

IV: Administer undiluted (20 mg/mL) as slow IV push over 1 to 2 minutes (Ref) maximum rate: 5 mg/minute (Ref).

Preparation for Administration: Pediatric

Parenteral: IV: Manufacturer does not recommend dilution. If further dilution is desired hydralazine should be diluted in NS for IVPB administration due to decreased stability in D5W; hydralazine diluted in NS to a concentration of 0.2 mg/mL is reported as stable for at least 1 day at room temperature (Ref).

Storage/Stability

Tablets: Store at 15°C to 30°C (59°F to 86°F).

Injection: Store at 20°C to 25°C (68°F to 77°F); use immediately after vial is opened. Solution may discolor upon contact with metal; discard discolored solutions.

Use: Labeled Indications

Hypertension: Management of moderate to severe hypertension.

Use: Off-Label: Adult

Heart failure with reduced ejection fraction; Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia)

Medication Safety Issues

Sound-alike/look-alike issues:

HyarALAZINE may be confused with hydroCHLORothiazide, HYDROmorphine, hydroOXYzine

Older Adult: High-Risk Medication:

Hyralazine is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with recurrent falls with postural hypotension or severe symptomatic aortic stenosis (O'Mahony 2023).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Dapoxetine: May increase orthostatic hypotensive effects of HyALAZINE. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease antihypertensive effects of HyALAZINE. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sildenafil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

Food Interactions

Food enhances bioavailability of hydralazine. Management: Administer without regard to food, but keep consistent.

Reproductive Considerations

Medications considered acceptable for the treatment of chronic hypertension during pregnancy may generally be used in patients trying to conceive. Patients planning to become pregnant may continue taking oral hydralazine for the treatment of chronic hypertension when effective (ACC/AHA [Whelton 2018]; NICE 2019; SOGC [Magee 2022]).

Pregnancy Considerations

Hydralazine crosses the placenta (Lamont 1986; Liedholm 1982).

High and frequent doses of IV hydralazine are associated with adverse events, such as maternal hypotension and fetal tachycardia, compared to other agents (ACOG 2019; ESC [Cífková 2020]; SOGC [Magee 2022]).

Chronic maternal hypertension is associated with adverse events in the fetus/infant. Chronic maternal hypertension may increase the risk of birth defects, low birth weight, premature delivery, stillbirth, and neonatal death. Actual fetal/neonatal risks may be related to the duration and severity of maternal hypertension. Untreated chronic hypertension may also increase the risks of adverse maternal outcomes, including gestational diabetes, preeclampsia, delivery complications, stroke, and myocardial infarction (ACOG 2019).

Due to pregnancy-induced physiologic changes and maternal acetylator status (*NAT2* genotype), some pharmacokinetic properties of oral hydralazine may be altered. Larger studies are needed to evaluate if or how these changes influence safety or efficacy during pregnancy (Han 2019).

Patients with preexisting hypertension may continue their medication during pregnancy unless contraindications exist (ESC [Regitz-Zagrosek 2018]). When treatment of hypertension is initiated during pregnancy, agents other than oral hydralazine may be preferred (ACOG 2019; ESC [Regitz-Zagrosek 2018]); oral hydralazine is considered an alternative option due to possible side effects (SOGC [Magee 2022]).

IV hydralazine is recommended for use in the management of acute onset, severe hypertension (systolic BP ≥ 160 mm Hg or diastolic BP ≥ 110 mm Hg) with preeclampsia or eclampsia in pregnant and postpartum patients (ACOG 2020; ESC [Cífková 2020]; SOGC [Magee 2022]).

Breastfeeding Considerations

Hydralazine is present in breast milk.

Data related to the presence of hydralazine in breast milk are available from a patient receiving oral hydralazine 50 mg 3 times a day, initiated prior to delivery. Hydralazine and metabolites were present in breast milk sampled 30 minutes after a midday dose at 8 weeks postpartum (Liedholm 1982). In a study of 4 breastfeeding women, breast milk concentrations of hydralazine were approximately one-half of maternal serum concentrations. Hydralazine was detected in the serum of their breastfeeding infants at a concentration 10 times smaller than concentrations found in a hypertensive infant being treated therapeutically with hydralazine (Lamont 1986).

The manufacturer recommends that caution be used if administered to patients who are breastfeeding. Hydralazine is considered compatible with breastfeeding; however, sufficient data are not available following long-term use (WHO 2002).

Monitoring Parameters

Blood pressure; heart rate; complete blood cell count (CBC); antinuclear antibody (ANA) titer.

Mechanism of Action

Direct vasodilation of arterioles (with little effect on veins) with decreased systemic resistance. Although exact mechanism unknown, arterial vasodilation may occur via inhibition of calcium release from the sarcoplasmic reticulum and inhibition of myosin phosphorylation in arterial smooth muscle cells (McComb 2016).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV: 10 to 80 minutes

Duration: IM, IV: Up to 12 hours (Marik 2007); **Note:** Duration may vary depending on acetylator status of patient. Hypotension due to hydralazine may last longer even though the circulating half-life is much shorter (Marik 2007; O'Malley 1975).

Absorption: Oral: Rapidly absorbed

Protein binding: 87%

Metabolism: Hepatically acetylated; extensive first-pass effect (oral)

Bioavailability: ~22% to 69% depending on acetylator status (increased in slow acetylators and decreased in rapid acetylators) (Ludden 1982); increased with food

Half-life elimination: 3 to 7 hours

Time to peak, plasma: Oral: 1 to 2 hours

Excretion: Urine (as metabolites)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Slow acetylators: Patients will generally have higher plasma levels of hydralazine and require lower doses to maintain control of BP.

Patient Counseling Points

(For additional information see "Hydralazine: Patient drug information")

What is this drug used for?

- It is used to treat high blood pressure. It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Headache

- Diarrhea
- Lack of appetite
- Vomiting
- Nausea

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Lupus like rash on the cheeks or other body parts, sunburn easy, muscle or joint pain, chest pain or shortness of breath, or swelling in the arms or legs
- Change in amount of urine passed
- Unable to pass urine
- Severe dizziness
- Passing out
- Chest pain
- Fast heartbeat
- Abnormal heartbeat
- Chills
- Sore throat
- Burning or numbness feeling
- Severe loss of strength and energy
- Bruising
- Bleeding
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Apo hyaralazine | Apresolin;
- (AR) Argentina: Hidral | Hydrapres;
- (AU) Australia: Alphapress | Apresoline;
- (BD) Bangladesh: Apresoline;
- (BR) Brazil: Apresolina;
- (CL) Chile: Hidralazina clorhidrato | Hidralazina Clorhidrato L.CH;
- (CN) China: Hdralzine hydrochloride;
- (CO) Colombia: Apresolina | Hidralazina clorhidrato;
- (CR) Costa Rica: Hidralazina clorhidrato;
- (DO) Dominican Republic: Apresolina | Hydrapres;
- (EC) Ecuador: Apresolina;
- (EG) Egypt: Ardiolazin;
- (ES) Spain: Hydrapres;
- (FI) Finland: Apresolin;
- (GB) United Kingdom: Apresoline | Hyaralazine | Hyaralazine aps | Hyaralazine berk | Hyaralazine cox | Hydralazine dc | Hyaralazine kent | Hyaralazine sandoz;
- (HK) Hong Kong: Apresoline;
- (ID) Indonesia: Apresoline;
- (IE) Ireland: Apresoline | Hyaralazine;
- (IL) Israel: Apresoline | Hyaralazine;
- (JO) Jordan: Apresoline | Barolex;
- (JP) Japan: Apresoline | Aprezine | Apudolomin | Asozart | Deselazin | Diucholin | Hypatol | Presfall | Propectin maruishi | Propectin organon | Solezarin | Tetrasoline | Travinson;
- (KE) Kenya: Apresoline | Hyaralazine;
- (KR) Korea, Republic of: Anprasol | Hydralazine | Samjin Hyaralazine HCl;
- (LT) Lithuania: Aprestin;
- (LV) Latvia: Aprestin;
- (MX) Mexico: Apresolina | Bionobal | Boguisina | Dilatec | Dinitryl | Folteden | Hidralazina | Hidralazina Mavi | Hidralazina solara;
- (MY) Malaysia: Apo hydralazine | Apresoline | Hyaralazine;
- (NG) Nigeria: Ak drazine | H zin | Hyaralazine | Timec hydralazine;
- (NL) Netherlands: Apresoline | Hyaralazine HCL;
- (NO) Norway: Apresolin | Hyaralazine | Hyaralazine accord;
- (NZ) New Zealand: Alphapress | Apresoline | Hyaralazine;
- (PA) Panama: Hidralazina | Hydrapres;
- (PH) Philippines: Apresoline | Aprezin | Hyperzine;
- (PK) Pakistan: Vaspro;
- (PR) Puerto Rico: Apresoline | Hydralazine | Hydralazine HCL;
- (PT) Portugal: Apresolina | Hidralazina;
- (QA) Qatar: Apresolina | Apresoline | Hydraz;
- (SA) Saudi Arabia: Apo hyaralazine | Apresolin | Aprolin | Hyaralazine;
- (SE) Sweden: Apresolin;
- (SG) Singapore: Apo hyaralazine | Apresoline | Hydralazine | Hydrallazine;
- (SL) Sierra Leone: H zin;
- (SR) Suriname: Apo hyaralazine | Apresoline | Hydralazine HCL;
- (TH) Thailand: Apresoline | Cesoline W | Cesoline y | Dilazin | Hilazin | Hydra | Hydralazin | Hyaralazine | Sanzine;
- (TR) Turkey: Apresolin;
- (TW) Taiwan: Adernan | Apolin | Apresoline | Aprezin | Apulon | Aslozine | Asozart | Diazide | Hyaralazine | Hylazine | Lotension | Naselin | Serpine | Solezarin | Stable | Susemide | Suyalin | Yajunlo;
- (UG) Uganda: Hyaralazine;
- (UY) Uruguay: Apresolina | Hidral;
- (VE) Venezuela, Bolivarian Republic of: Apresolina;
- (ZA) South Africa: Apresoline | Hyaralazine | Hyperphen | Rolab-hyaralazine | Sandoz hyaralazine hcl;

- (ZM) Zambia: Hyaralazine HCL;
- (ZW) Zimbabwe: Apo hyaralazine | Hyperphen

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